

Home Task - 3

Student Name: Bharti Mehra
Branch: CSE - AIML
Semester: 5
Subject Name: ADBMS

UID: 24BAI70051
Section/Group: 24AIT_KRG-1_B
Date of Performance: 27/7/26
Subject Code: 24CSP-310

Q1.

Given a table containing information about bank deposits and withdrawals made using Paypal, write a query to retrieve the final account balance for each account, taking into account all the transactions recorded in the table with the assumption that there are no missing transactions.

transactions Table:

Column Name	Type
transaction_id	integer
account_id	integer
amount	decimal
transaction_type	varchar

transactions Example Input:

transaction_id	account_id	amount	transaction_type
123	101	10.00	Deposit
124	101	20.00	Deposit
125	101	5.00	Withdrawal
126	201	20.00	Deposit
128	201	10.00	Withdrawal

Example Output:

account_id	final_balance
101	25.00
201	10.00

Solution:

```
SELECT ACCOUNT_ID,  
SUM(  
  CASE  
    WHEN TRANSACTION_TYPE='Deposit' THEN AMOUNT  
    WHEN TRANSACTION_TYPE='Withdrawal' THEN -AMOUNT  
    ELSE 0  
  END  
) AS FINAL_BALANCE  
FROM TRANSACTIONS  
GROUP BY ACCOUNT_ID;
```